

Problem 27.10 Eq.(22.1.11)

$$u(\lambda, T)d\lambda = \frac{8\pi}{\lambda^4} \left(\frac{hc}{\lambda} \right) \frac{1}{e^{\beta hc/\lambda} - 1} d\lambda$$

Set $\beta hc / \lambda = x; \quad \lambda T = hc/k_B x$

$$\frac{d}{dx} \left(\frac{x^5}{e^x - 1} \right) = 0$$

$$5 - x = 5e^{-x}; \quad x = 4.965; \quad \lambda_m T = 2898 \mu\text{m} \cdot \text{K}$$

Problem 27.10 Eq.(22.1.10)

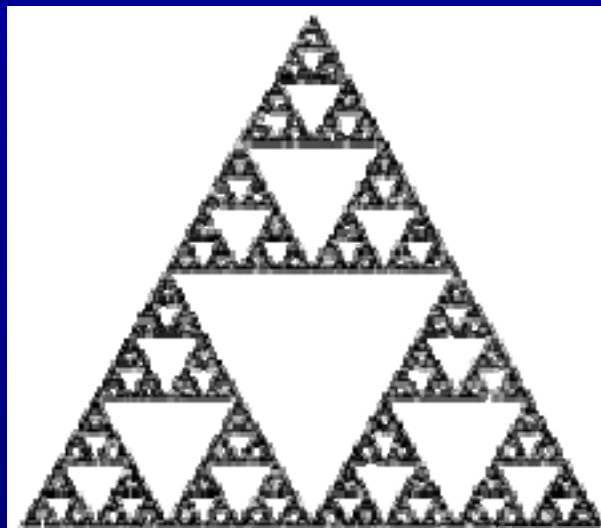
$$u(\nu, T) d\nu = \frac{8\pi h}{c^3} \frac{\nu^3}{e^{\beta h \nu} - 1} d\nu$$

Set $\beta h \nu = x; \quad \frac{h \nu}{k_B T} = x$

$$\frac{d}{dx} \left(\frac{x^3}{e^x - 1} \right) = 0$$

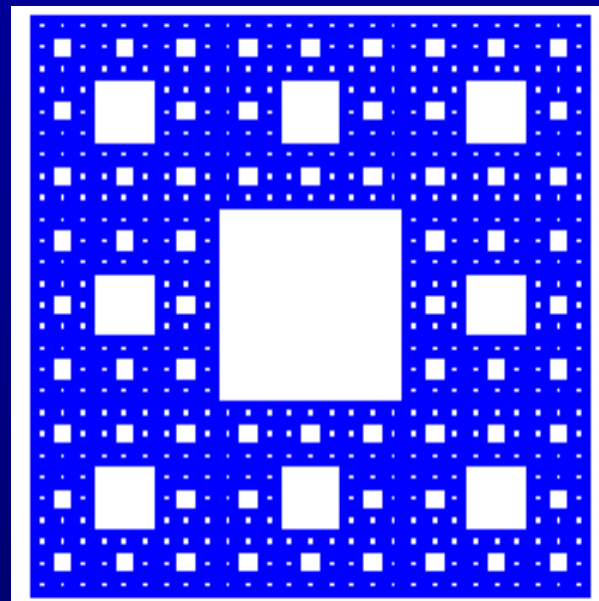
$$3 - x = 3e^{-x}; \quad x = 2.822; \quad \frac{h \nu_m}{k_B T} = 2.822$$

Problem 28.2



$$D_s = \frac{\ln n}{\ln m} = \frac{\ln 3}{\ln 2} \\ = 1.58496 \dots$$

The Sierpinski gasket



$$D_s = \frac{\ln n}{\ln m} = \frac{\ln 8}{\ln 3} \\ = 1.89276 \dots$$

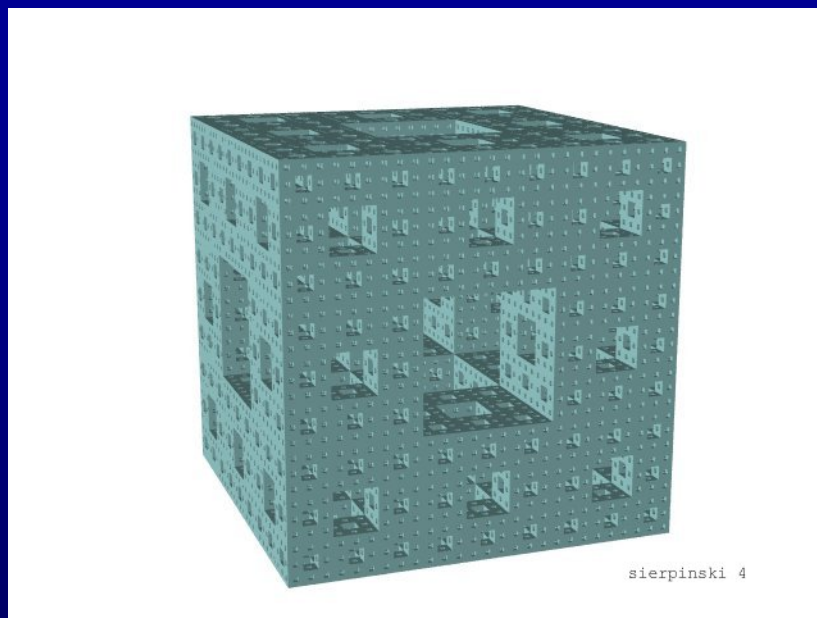
Similarity dimension

When a shape is composed of n similar shape of size $1/m$, then the similarity dimension is given by

$$D_s = \frac{\ln n}{\ln m}.$$

Problem 28.3

The Sierpinski sponge



$$D_s = \frac{\ln n}{\ln m} = \frac{\ln 20}{\ln 3} = 2.72683\dots$$

Problem 28.7(a) (b)

$$(a) \quad U = U_0 - \mu - eEx - \frac{e_s^2}{4x}$$

$$\frac{\partial U}{\partial x} = 0 = -eE + \frac{e_s^2}{4x^2}; \quad x = \frac{e_s}{2(eE)^{1/2}};$$

$$\frac{\partial^2 U}{\partial x^2} < 0$$

$$\begin{aligned} U_{\max} &= U_0 - \mu - \frac{1}{2}(eE)^{1/2} e_s - \frac{e_s}{4} \times 2(eE)^{1/2} \\ &= U_0 - \mu - (eE)^{1/2} e_s \end{aligned}$$

$$(b) \quad e_s^2 \equiv \frac{e^2}{4\pi\epsilon_0} \equiv k_0 e^2; \quad e_s = k_0^{1/2} e;$$

$$\begin{aligned} U_{\max} &= W - k_0^{1/2} e (eE)^{1/2} \\ &= W - k_0^{1/2} e^{3/2} E^{1/2} \end{aligned}$$

Replace W in Eq.(28.4.3) with $U_{\max}$ and we have (no quantum tunneling considered):

$$J_x = \frac{4\pi m}{h^3} (k_B T)^2 \exp\left(-\frac{W}{k_B T}\right) \exp\left(\frac{k_0^{1/2} e^{3/2} E^{1/2}}{k_B T}\right)$$

Problem 29.7 The total binding energy is (atomic masses are used; the masses of electrons are canceled out)

$$\begin{aligned} BE &= Nm_n c^2 + Zm_p c^2 - m_{\text{Pu}} c^2 \\ &= Nm_n c^2 + Zm[{}^1\text{H}]c^2 - m[{}^{239}\text{Pu}]c^2 \\ &= (145 \times 1.00876 + 94 \times 1.00783 \\ &\quad - 239.05216) \times 931.5 \text{ MeV} \\ &= 1820.21 \text{ MeV} \\ be &= 7.6 \text{ MeV} \end{aligned}$$

Problem 29.12

$$5\text{MeV} \times \frac{k_{\text{B}} 300\text{K}}{\frac{1}{40}\text{eV}} = k_{\text{B}} 6 \times 10^{10} \text{K}$$

Problem 29.14

$$\frac{N}{N_0} = e^{-t \ln 2 / t_{1/2}} = \left(\frac{1}{2} \right)^{50.0/12.3} = 5.97\%$$

Problem 29.23



$$\begin{aligned} Q &= m_{\text{Be}}c^2 - m_{\text{B}}c^2 - m_e c^2 \\ &= (11.021660 - 11.009305) \times 931.5 \text{MeV} - 0.511 \text{MeV} \\ &= 11.0 \text{MeV} \end{aligned}$$

Problem 29.27

$$R_0 = 12.4 / \text{min}, R(t) = 3.5 / \text{min}$$

$$\begin{aligned} t &= \frac{\ln[R_0/R(t)]}{\ln 2} t_{1/2} \\ &= \frac{\ln(12.4/3.5)}{\ln 2} \times 5.57 \times 10^3 \text{ a} \\ &= 10.2 \times 10^3 \text{ a} \end{aligned}$$